

Администрирование сетевых подсистем

Настройка DHCP-сервера (Лабораторная работа №3)

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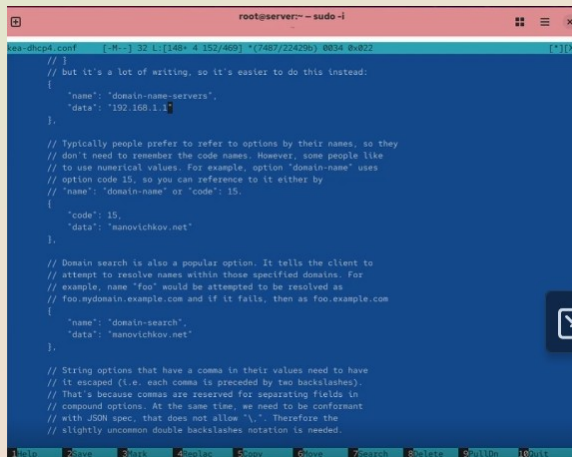
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Выполнение лабораторной работы

Конфигурирование DHCP-сервера



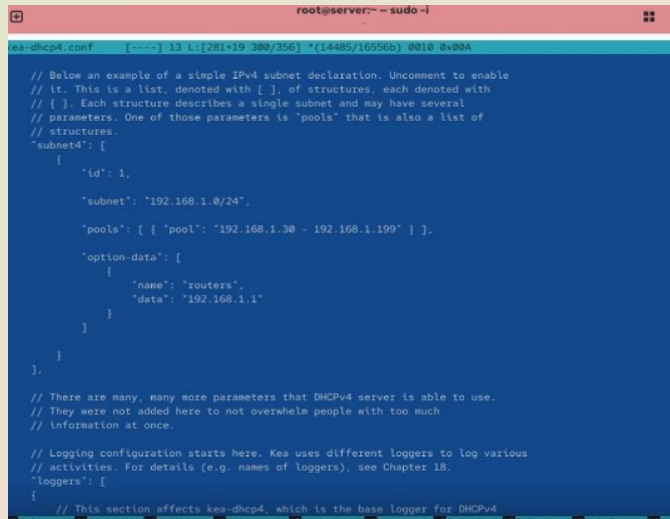
```
root@server:~# sudo -i
keahelp4.conf [-M--] 32 L:[148+ 4 152/469] *(7487/224296) 0034 0x022 [*][X]
// }
// but it's a lot of writing, so it's easier to do this instead:
{
  "name": "domain-name-servers",
  "data": "192.168.1.1"
},

// Typically people prefer to refer to options by their names, so they
// don't need to remember the code names. However, some people like
// to use numerical values. For example, option "domain-name" uses
// option code 15, so you can reference to it either by
// "name": "domain-name" or "code": 15.
{
  "code": 15,
  "data": "manovichkov.net"
},

// Domain search is also a popular option. It tells the client to
// attempt to resolve names within those specified domains. For
// example, name "foo" would be attempted to be resolved as
// foo.mydomain.example.com and if it fails, then as foo.example.com
{
  "name": "domain-search",
  "data": "manovichkov.net"
},

// String options that have a comma in their values need to have
// it escaped (i.e. each comma is preceded by two backslashes).
// That's because commas are reserved for separating fields in
// compound options. At the same time, we need to be conformant
// with JSON spec, that does not allow "\". Therefore the
// slightly uncommon double backslashes notation is needed.
```

Рис. 1: Редактирование domain-name и domain-search



```
root@server:~ -- sudo -i
kea-dhcp4.conf  [----] 13 L:[281+19 300/356] *(14485/16556b) 0010 0x00A

// Below an example of a simple IPv4 subnet declaration. Uncomment to enable
// it. This is a list, denoted with [ ], of structures, each denoted with
// { }. Each structure describes a single subnet and may have several
// parameters. One of those parameters is "pools" that is also a list of
// structures.
"subnet4": [
  {
    "id": 1,

    "subnet": "192.168.1.0/24",

    "pools": [ { "pool": "192.168.1.30 - 192.168.1.199" } ],

    "option-data": [
      {
        "name": "routers",
        "data": "192.168.1.1"
      }
    ]
  }
]

// There are many, many more parameters that DHCPv4 server is able to use.
// They were not added here to not overwhelm people with too much
// information at once.

// Logging configuration starts here. Kea uses different loggers to log various
// activities. For details (e.g. names of loggers), see Chapter 18.
"loggers": [
  {
    // This section affects kea-dhcp4, which is the base logger for DHCPv4
```

Рис. 2: Конфигурация подсети DHCP

```
manovichkov.net  [-----] 0 L:[ 1+13 14/ 14] *(244 / 244b) <EOF>
$TTL 1D
@<----->IN SOA<@ server.manovichkov.net. (
<-----><-----><-----><-----><-----><----->2025091700<----->; serial
<-----><-----><-----><-----><-----><----->1D<----->; refresh
<-----><-----><-----><-----><-----><----->1H<----->; retry
<-----><-----><-----><-----><-----><----->1W<----->; expire
<-----><-----><-----><-----><-----><----->3H )<--->; minimum
<----->NS<----->@
<----->A<----->192.168.1.1
$ORIGIN manovichkov.net.
server<A<----->192.168.1.1
ns<---->A<----->192.168.1.1
dhcp<-->A<----->192.168.1.1
```

Рис. 3: Редактирование прямой зоны DNS

```
192.168.1      [-M--] 29 L:[ 1+13 14/ 15] *(298 / 299b) 0010 0x00A
$TTL 1D
@<----->IN SOA<@ server.manovichkov.net. (
<-----><-----><-----><-----><----->2025091700<----->; serial
<-----><-----><-----><-----><----->1D<----->; refresh
<-----><-----><-----><-----><----->1H<----->; retry
<-----><-----><-----><-----><----->1W<----->; expire
<-----><-----><-----><-----><----->3H )<----->; minimum
<----->NS<----->@
<----->A<----->192.168.1.1
<----->PTR<---->server.manovichkov.net.
$ORIGIN 1.168.192.in-addr.arpa.
1<----->PTR<---->server.manovichkov.net.
1<----->PTR<---->ns.manovichkov.net.
1<----->PTR<---->dhcp.manovich
```

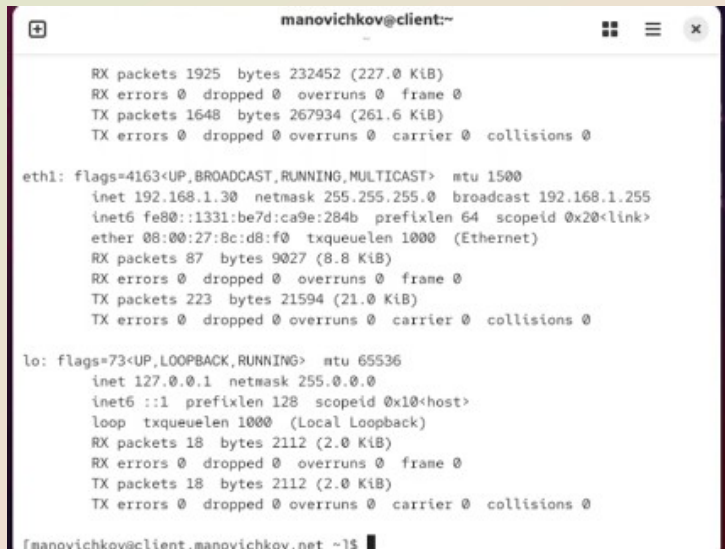
Рис. 4: Редактирование обратной зоны DNS

Проверка работы DHCP

```
[root@server.manovichkov.net ~]# systemctl restart named
[root@server.manovichkov.net ~]# ping dhcp.manovichkov.net
PING dhcp.manovichkov.net (192.168.1.1) 56(84) bytes of data.
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=1 ttl=64 time=0.017 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=2 ttl=64 time=0.093 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=3 ttl=64 time=0.096 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=4 ttl=64 time=0.080 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=5 ttl=64 time=0.080 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=6 ttl=64 time=0.038 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=7 ttl=64 time=0.063 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=8 ttl=64 time=0.024 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=9 ttl=64 time=0.212 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=10 ttl=64 time=0.084 ms
64 bytes from server.manovichkov.net (192.168.1.1): icmp_seq=11 ttl=64 time=0.092 ms
```

Рис. 5: Проверка доступности DHCP-сервера

Анализ работы DHCP-сервера



```
manovichkov@client:~  
  
RX packets 1925  bytes 232452 (227.0 KiB)  
RX errors 0  dropped 0  overruns 0  frame 0  
TX packets 1648  bytes 267934 (261.6 KiB)  
TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0  
  
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500  
    inet 192.168.1.30  netmask 255.255.255.0  broadcast 192.168.1.255  
    inet6 fe80::1331:be7d:ca9e:284b  prefixlen 64  scopeid 0x20<link>  
    ether 08:00:27:8c:d8:f0  txqueuelen 1000  (Ethernet)  
    RX packets 87  bytes 9027 (8.8 KiB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 223  bytes 21594 (21.0 KiB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536  
    inet 127.0.0.1  netmask 255.0.0.0  
    inet6 ::1  prefixlen 128  scopeid 0x10<host>  
    loop txqueuelen 1000  (Local Loopback)  
    RX packets 18  bytes 2112 (2.0 KiB)  
    RX errors 0  dropped 0  overruns 0  frame 0  
    TX packets 18  bytes 2112 (2.0 KiB)  
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0  
  
[manovichkov@client.manovichkov.net ~]$
```

Рис. 6:Сетевые интерфейсы на клиенте

```
[root@server.manovichkov.net ~]#  
[root@server.manovichkov.net ~]# cat /var/lib/kea/kea-leases4.csv  
address,hwaddr,client_id,valid_lifetime,expire,subnet_id,fqdn_fwd,fqdn_rev,hostname,state,user_context,pool_id  
192.168.1.30,08:00:27:8c:d8:f0,01:08:00:27:8c:d8:f0,3600,1758097128,1,0,0,client,0,,0  
192.168.1.30,08:00:27:8c:d8:f0,01:08:00:27:8c:d8:f0,3600,1758097129,1,0,0,client,0,,0  
192.168.1.30,08:00:27:8c:d8:f0,01:08:00:27:8c:d8:f0,3600,1758097134,1,0,0,client,0,,0  
[root@server.manovichkov.net ~]#
```

Рис. 7: Файл с арендой DHCP-адресов

Настройка обновления DNS-зоны

```
// Note: empty-zones-enable yes; option is default.
// If private ranges should be forwarded, add.
// disable-empty-zone "."; into options
//.

zone "manovichkov.net" IN {
    <----->type master;
    <----->file "master/fz/manovichkov.net";
    <----->update-policy {
    <----->    grant DHCP_UPDATER wildcard *.manovichkov.net A DHCID;
    <----->};
};

zone "1.168.192.in-addr.arpa" IN {
    <----->type master;
    <----->file "master/rz/192.168.1";
    <----->update-policy {
    <----->    grant DHCP_UPDATER wildcard *.168.192.in-addr.arpa PTR DHCID;
    <----->};
};
```

Рис. 8: Настройка зон с поддержкой обновления

```
tsig-keys.json  [----] 2 L:[ 1+ 6 7/ 8] *(177 / 178b) 0010 0x00A [*][X]
"tsig-keys": [
{
<----->"name": "DHCP_UPDATER",
<----->"algorithm": "hmac-sha512",
<----->"secret": "+K+UYj2Zk0gramR5lAHopZrk7P67skfE3yA6/YH2eAv7PGx9DoW+jzrqPMFmvOBw7mcmOPUn9Zl6FrF2mPOUfQ=="
}
],
```

Рис. 9: Файл tsig-keys.json

```
root@server:~# sudo -i
kea-dhcp-ddns.conf [-M-] 4 L:[ 33*23 56/ 99] *(1823/3402b) 0111 0x06f [X]
//-----
// "name": "manovichkov.net.",
// "key-name": "DHCP_UPDATER",
// "dns-servers": [
//     { "ip-address": "192.168.1.1" }
// ]
// }
// },
// "reverse-ddns" : {
//     "ddns-domains": [
//     {
//         "name": "1.168.192.in-addr.arpa.",
//         "key-name": "DHCP_UPDATER",
//         "dns-servers": [
//             { "ip-address": "192.168.1.1" }
//         ]
//     }
//     ]
// },
// Logging configuration starts here. Kea uses different loggers to log various
// activities. For details (e.g. names of loggers), see Chapter 18.
"loggers": [
{
    // This specifies the logging for D2 (DHCP-DDNS) daemon.
    "name": "kea-dhcp-ddns",
    "output-options": [
    {
        // Specifies the output file. There are several special values
        // supported:
        // - stdout (prints on standard output)
        // - stderr (prints on standard error)
```

Проверка работы Kea DDNS

```
[root@server.manovichkov.net ~]# chown kea:kea /etc/kea/kea-dhcp-ddns.conf
[root@server.manovichkov.net ~]# kea-dhcp-ddns -t /etc/kea/kea-dhcp-ddns.conf
2025-09-17 07:39:49.575 INFO [kea-dhcp-ddns.dctl/19493.140431892119872] DCTL_CONFIG_CHECK_COMPLETE server has completed configuration check: listening on 127.0.0.1, port 53001, using UDP, result: success(0), text=Configuration check successful
[root@server.manovichkov.net ~]# systemctl enable --now kea-dhcp-ddns.service
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp-ddns.service' -> '/usr/lib/systemd/system/kea-dhcp-ddns.service'.
[root@server.manovichkov.net ~]# systemctl status kea-dhcp-ddns.service
● kea-dhcp-ddns.service - Kea DHCP-DDNS Server
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp-ddns.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-09-17 07:40:18 UTC; 11s ago
 Invocation: 6b1f526e5af04486bb5cb220a77b0f0b
    Docs: man:kea-dhcp-ddns(8)
 Main PID: 19711 (kea-dhcp-ddns)
   Tasks: 5 (limit: 10381)
  Memory: 1.7M (peak: 6M)
     CPU: 10ms
  CGroup: /system.slice/kea-dhcp-ddns.service
          └─19711 /usr/sbin/kea-dhcp-ddns -c /etc/kea/kea-dhcp-ddns.conf

Sep 17 07:40:18 server.manovichkov.net systemd[1]: Started kea-dhcp-ddns.service - Kea DHCP-DDNS Server.
Sep 17 07:40:18 server.manovichkov.net kea-dhcp-ddns[19711]: 2025-09-17 07:40:18.896 INFO [kea-dhcp-ddns.dctl/19
Sep 17 07:40:18 server.manovichkov.net kea-dhcp-ddns[19711]: INFO COMMAND_ACCEPTOR_START Starting to accept conn
Sep 17 07:40:18 server.manovichkov.net kea-dhcp-ddns[19711]: INFO DCTL_CONFIG_COMPLETE server has completed conf
Sep 17 07:40:18 server.manovichkov.net kea-dhcp-ddns[19711]: INFO DHCP_DDNS_STARTED Kea DHCP-DDNS server version
lines 1-17/17 (END)
```

Рис. 11:Запуск службы kea-dhcp-ddns

```
"Dhcp4": {  
  // Add names of your network interfaces to listen on.  
  "interfaces-config": {  
    // See section 8.2.4 for more details. You probably want to add just  
    // interface name (e.g. "eth0" or specific IPv4 address on that  
    // interface name (e.g. "eth0/192.0.2.1").  
    "interfaces": [ "eth1" ]  
  
    // Kea DHCPv4 server by default listens using raw sockets. This ensures  
    // all packets, including those sent by directly connected clients  
    // that don't have IPv4 address yet, are received. However, if your  
    // traffic is always relayed, it is often better to use regular  
    // UDP sockets. If you want to do that, uncomment this line:  
    // "dhcp-socket-type": "udp"  
  },  
  
  "dhcp-ddns": {  
    <----->"enable-updates": true  
  },  
  ....  
  "ddns-qualifying-suffix": "manovichkov.net",  
  "ddns-override-client-update": true  
  ....
```

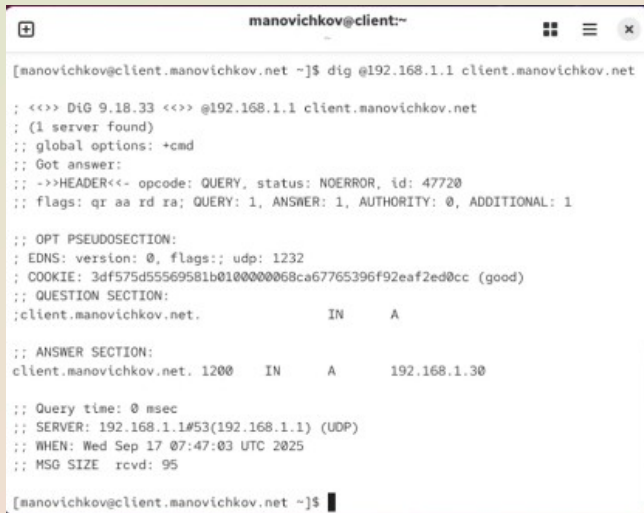
Рис. 12:Изменения в конфигурации kea-dhcp4.conf

Перезапуск DHCP-сервера

```
[root@server.manovichkov.net ~]#  
[root@server.manovichkov.net ~]# mcedit /etc/kea/kea-dhcp4.conf  
  
[root@server.manovichkov.net ~]# systemctl restart kea-dhcp4.service  
[root@server.manovichkov.net ~]# systemctl status kea-dhcp4.service  
● kea-dhcp4.service - Kea DHCPv4 Server  
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)  
   Active: active (running) since Wed 2025-09-17 07:45:55 UTC; 2s ago  
  Invocation: a12027e44f9241409826b74b3b8bd2fe  
     Docs: man:kea-dhcp4(8)  
    Main PID: 20508 (kea-dhcp4)  
      Tasks: 7 (limit: 10381)  
     Memory: 2.5M (peak: 6M)  
        CPU: 13ms  
    CGroup: /system.slice/kea-dhcp4.service  
            └─20508 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf  
  
Sep 17 07:45:55 server.manovichkov.net systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.  
Sep 17 07:45:55 server.manovichkov.net kea-dhcp4[20508]: 2025-09-17 07:45:55.809 INFO [kea-dhcp4.dhcp4/20508.1408  
Sep 17 07:45:55 server.manovichkov.net kea-dhcp4[20508]: 2025-09-17 07:45:55.809 INFO [kea-dhcp4.commands/20508.8  
lines 1-15/15 (END)
```

Рис. 13: Запуск DHCP-сервера после изменений

Проверка DNS-записи клиента



```
manovichkov@client:~  
[manovichkov@client.manovichkov.net ~]$ dig @192.168.1.1 client.manovichkov.net  
  
; <<>> DiG 9.18.33 <<>> @192.168.1.1 client.manovichkov.net  
; (1 server found)  
;; global options: +cmd  
;; Got answer:  
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 47720  
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 1232  
; COOKIE: 3df575d55569581b0100000068ca67765396f92eaf2ed0cc (good)  
;; QUESTION SECTION:  
client.manovichkov.net.          IN      A  
  
;; ANSWER SECTION:  
client.manovichkov.net. 1200     IN      A      192.168.1.30  
  
;; Query time: 0 msec  
;; SERVER: 192.168.1.1#53(192.168.1.1) (UDP)  
;; WHEN: Wed Sep 17 07:47:03 UTC 2025  
;; MSG SIZE rcvd: 95  
  
[manovichkov@client.manovichkov.net ~]$
```

Рис. 14: Проверка DNS-записи клиента

Выводы по проделанной работе

В ходе работы была развернута инфраструктура с двумя виртуальными машинами — `server` и `client`.

На сервере установлены и настроены службы `DNS` и `DHCP`, реализована поддержка `DDNS`.

Клиенты автоматически получают IP-адреса и регистрируются в DNS-зоне.

Конфигурации вынесены в окружение Vagrant для автоматизации запуска.